Rectangular matrix multiplication from polynomial products

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Abstract

In this note, we extend the matrix-multiplication analysis in [9] to rectangular products and prove

$$\omega(1,k,1) \le \begin{cases} 2, & 0 \le k \le \frac{1}{2}, \\ 1+k+\frac{1}{4k}, & k \ge \frac{1}{2}. \end{cases}$$

In particular, $\omega(1,\frac{1}{2},1) = 2$ and the dual exponent satisfies $\alpha \ge \frac{1}{2}$. The proof uses explicit degenerations of convolution tensors, multinomial counting, and Strassen's spectral theorem. An angular constraint on three dot-product parameters gives both the rectangular curve and a bound for $\omega(a,b,c)$, with exact exponent $a+b$ whenever $a \ge b \ge c \ge 0$ and $c \le \frac{ab}{a+b}$. As an application, Zwick's algorithm [15] for all-pairs shortest paths in directed unweighted graphs runs in $O(n^{2.5})$ time. Combining the rectangular bound with the $(\min,+)$-product improvement of Alman and Vassilevska Williams [1] further gives $O(n^{2.4999})$ running time.

1. Introduction

Disclaimer. This note, its proofs, analysis and writeup were obtained fully through OpenAI Codex (GPT-6.1). The author has read and analyzed this note in full and takes full responsibility for its contents, and takes no credit for this work.

The initial motivation for this note was to explore whether the methods of the groundbreaking matrix-multiplication result in [9] could be pushed to analyze rectangular matrix products. We also aim to gather useful bounds and their algorithmic consequences in one place. Rephrasing the proofs in a form accessible to the theoretical computer science community has value in its own right.

Complexity convention. Throughout this note, all complexity bounds suppress a $+o(1)$ term in the exponent: $O(n^c)$ denotes $n^{c+o(1)}$. All shape parameters are fixed.

Rectangular matrix multiplication is a basic ingredient in algorithms for all-pairs problems, including shortest paths, lowest common ancestors, and Hamming distance. Its exponents depend on the shape of the product; an improvement for square matrices need not give a comparable improvement for every rectangular shape. The analysis in [9] establishes the square bound $\omega \le 2.25$ using a shared-leg entropy inequality and degenerations of convolution tensors. We extend that analysis to rectangular products by retaining the orientation information lost under full leg symmetrization.

We work over $\mathbb{C}$. The resulting exponent bounds hold over every field of characteristic zero by invariance of asymptotic tensor rank under field extension [10, Theorem 3.10].

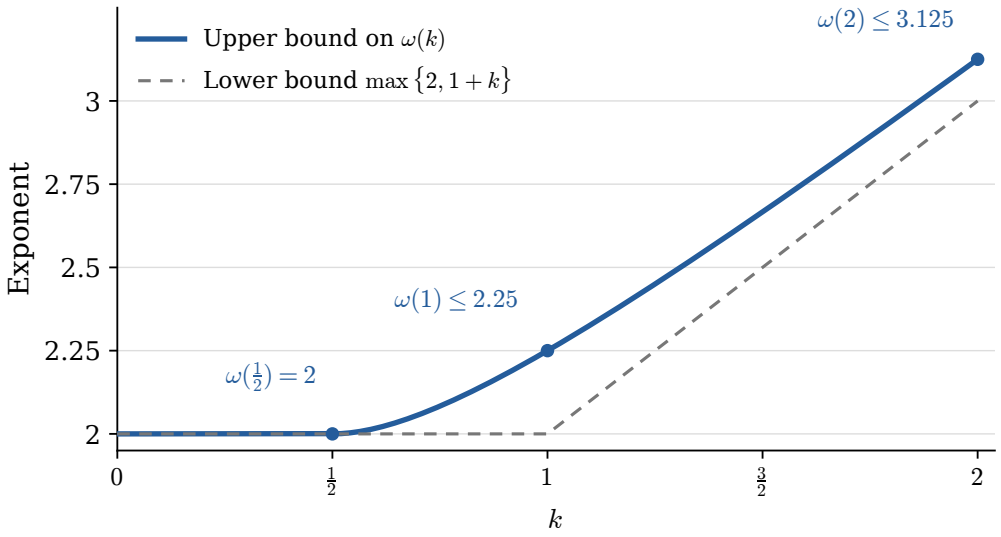


Figure 1: The upper bound on $\omega(k)$ from Corollary 1.2 for $0 \le k \le 2$, together with the size lower bound $\max\{2, 1+k\}$. The bounds coincide for $0 \le k \le \frac{1}{2}$.

For fixed nonnegative real u, v, w, let $\omega(u, v, w)$ be the infimum of the exponents e for which multiplying an $\lceil n^u \rceil \times \lceil n^v \rceil$ matrix by a $\lceil n^v \rceil \times \lceil n^w \rceil$ matrix uses $O(n^e)$ arithmetic operations, with the convention above. Write $\omega(k) = \omega(1, k, 1)$ and $\alpha = \sup\{k \ge 0 : \omega(k) = 2\}$.

We first state the bound for general matrix shapes, with the exponents sorted in decreasing order. We interpret the threshold $\frac{ab}{a+b}$ as zero when $a = b = 0$.

Theorem 1.1. *For every fixed $a \ge b \ge c \ge 0$,*

$$\omega(a,b,c) \le \begin{cases} a+b, & c \le \dfrac{ab}{a+b}, \\ \dfrac{a+b+c}{2} + \dfrac{1}{4}\left(\dfrac{ab}{c} + \dfrac{ac}{b} + \dfrac{bc}{a}\right), & c > \dfrac{ab}{a+b}. \end{cases} \tag{1}$$

When $c \le \frac{ab}{a+b}$, equality holds by the size lower bound.

Applying this bound to the sorted dimensions of $(1, k, 1)$ gives the rectangular curve.

Corollary 1.2. *For every fixed real $k \ge 0$,*

$$\omega(k) \le \begin{cases} 2, & 0 \le k \le \frac{1}{2}, \\ 1 + k + \dfrac{1}{4k}, & k \ge \frac{1}{2}. \end{cases} \tag{2}$$

Consequently $\omega(k) = 2$ for $0 \le k \le \frac{1}{2}$.

Corollary 1.2 gives exponent two for inner dimension up to $n^{0.5}$, and an upper bound within $\frac{1}{4k}$ of the input-size lower bound $1 + k$ for $k \ge 1$.

The proof has two parts. First, we list explicit ways to split and combine polynomial products and apply a tensor character to them. A character assigns numerical sizes that add on direct sums, multiply on tensor products, and cannot increase under restriction. Multinomial counting gives concavity and growth inequalities for these sizes. Rescaling, common slopes, and affine bounds then give $\sum_i \arccos \sqrt{p_i} \ge \pi/2$ for the character's three dot-product parameters.

Second, a squared-norm identity for triangle normals bounds weighted sums of these parameters. We apply Strassen's spectral theorem: the largest character value equals asymptotic tensor rank. We state precisely the form of this theorem we use, then show how the rank bound gives a running-time bound for every fixed matrix shape.

Relation to the preceding analysis. The two polynomial splittings in Lemma 2.1 recap the proofs of Lemmas 4.1 and 4.2 of [9] in coefficient coordinates. The finite separation adapts Proposition 3.1 of that reference. All three constructions are proved in the appendices. Multinomial counting then gives the required power-sum inequality. The replication construction, rescaling, and coupling of six affine offsets supply the rectangular extension. The weighted bound is proved by an elementary identity for triangle normals. Dropping its nonnegative squared-norm term recovers the classical Wolstenholme inequality [8, Equation (1)]. Strassen's spectral theorem is a standard input; a modern exposition is [12]. We cite it rather than reproduce its proof. We make no priority claim for the standard tools or the finite construction variants. Section 7 substitutes the bounds into known algorithmic tradeoffs.

2. Polynomial products and finite constructions

We begin by describing matrix multiplication and convolution as tensors. We then build ways to split polynomial products and separate computations that share variables. These constructions give the inequalities in Section 3. A further construction replicates a convolution; it will let us compare convolutions at different scales in Section 4.

A tensor is a trilinear form in three groups of variables, called its *legs*. A *restriction* makes linear substitutions independently on the three legs. A *degeneration* is a limit of such restrictions: the substitutions may depend on $\varepsilon \neq 0$, and every resulting coefficient converges as $\varepsilon \to 0$. Our constructions first change bases and then multiply variables by integer powers of ε. The resulting coefficient array is a polynomial in ε whose constant coefficient is the desired tensor. We write $T \geq S$ for a restriction and $T \trianglerighteq S$ for a degeneration. Successive constructions can use a single parameter: replace the earlier parameter by a sufficiently large power of the later one, so every earlier error term still has positive total exponent.

The $\mathrm{rank}(T)$ is the least number of products of three linear forms, one per leg, summing to T. The direct sum $S \oplus T$ uses disjoint variables on all three legs and represents independent computations. The tensor product $S \otimes T$ pairs their indices, as in a Kronecker product, and represents their composition on blocks. We write mT for m independent copies, and $T^{\otimes N}$ for N tensor factors. Thus $\mathrm{rank}(T)\, xyz \geq T$. All factors used below restrict to xyz by keeping one nonzero term; this is what it means to drop a factor.

The oriented dot products are

$$B_X(q) = x \sum_{j=1}^{q} y_j z_j,$$

with $B_Y(q), B_Z(q)$ obtained by permuting the legs. They have rank q, and $B_i(q) \otimes B_i(q')$ is $B_i(qq')$ after reindexing. The matrix multiplication tensor factors as

$$\langle A, B, C \rangle = \sum_{u,v,w} x_{uv} y_{vw} z_{wu} \cong B_Y(A) \otimes B_Z(B) \otimes B_X(C). \tag{3}$$

Thus the three dot-product dimensions specify the matrix shape.

For distinct input legs i, j, with remaining output leg k, the *convolution tensor* is

$$C_{ij}(r, s) = \sum_{h=0}^{r-1} \sum_{\ell=0}^{s-1} x_h^{(i)} x_\ell^{(j)} x_{h+\ell}^{(k)}.$$

It multiplies two polynomials represented by their coefficient vectors. Its rank is $r+s-1$: evaluation at $r+s-1$ points and interpolation give the upper bound, and the independent output coefficients give the matching matrix-rank lower bound. Decreasing either input length is a restriction. Also

$$C_{ij}(1,s) \cong B_i(s), \qquad C_{ij}(r,1) \cong B_j(r). \tag{4}$$

Blocks T_h are *matched sectors sharing leg i* when $T = \sum_h T_h$ and the blocks may share variables on leg i, but use pairwise disjoint variables on each other leg. The following constructions hold for every permutation of the legs.

Lemma 2.1 (Finite separation and polynomial splittings)**.**

1. *If T has M matched sectors sharing leg i, then*

$$(2M)T \trianglerighteq \bigoplus_{h=1}^{M} (T_h \otimes B_i(M)). \tag{5}$$

   The sectors on the right are independent on all three legs.

2. *For $r \geq 1$ and $s \geq 2$, $C_{ij}(r,s) \otimes B_i(2)$ degenerates to two matched sectors sharing leg i, namely $C_{ij}(r,s+1)$ and $C_{ij}(r,s-1)$.*

3. *For $r,s \geq 1$, $C_{ij}(r,3s+r-1)$ degenerates to three matched sectors sharing leg i: two copies of $C_{ij}(r,s)$ and one copy of $C_{ik}(r,s)$.*

Appendices A and B give the polynomial splittings in coefficient coordinates. Appendix C proves (5) using a finite geometric sum and a nonnegative square. The next construction splits a convolution into repeated independent copies of a smaller convolution.

Lemma 2.2 (Replicating a convolution)**.** *For all positive integers m, r, s,*

$$C_{ij}(mr, ms) \trianglerighteq mC_{ij}(r,s). \tag{6}$$

The construction combines polynomial division with the standard evaluation description of cyclic convolution [5, Theorem 5 and Section 5.1]. We give the full proof in Appendix E.

3. Tensor characters and counting

To use the preceding constructions, we assign numerical values to tensors using tensor characters. The splittings and separation give concavity and tripling inequalities for their convolution values, which we rescale in the next section. We also identify the three parameters that determine their matrix-multiplication values and state Strassen's theorem: bounding every character value will give an asymptotic rank bound in Section 6.

A *tensor character* is a function λ from tensors to nonnegative real numbers satisfying

$$\begin{aligned} &\lambda(S \oplus T) = \lambda(S) + \lambda(T), \quad \lambda(S \otimes T) = \lambda(S)\lambda(T), \\ &\lambda(xyz) = 1, \qquad\qquad\qquad S \geq T \implies \lambda(S) \geq \lambda(T). \end{aligned} \tag{7}$$

These rules express addition of independent computations, multiplication of tensor factors, and monotonicity under linear substitutions. Matrix rank after grouping one leg against the other two is an example. Keeping one nonzero term and using $\operatorname{rank}(T)\, xyz \geq T$ gives $1 \leq \lambda(T) \leq \operatorname{rank}(T)$ for every nonzero T.

Lemma 3.1 (Interpolation of a degeneration)**.** *Suppose a degeneration $T \trianglerighteq S$ produces a tensor $P(\varepsilon)$ whose coefficients are polynomials of degree at most L. Then, for every integer $N \geq 1$,*

$$(NL+1)T^{\otimes N} \geq S^{\otimes N}. \tag{8}$$

Consequently, every tensor character satisfies $\lambda(S) \leq \lambda(T)$.

Equation (8) is the standard interpolation conversion from degeneration to restriction [2, Theorem 8], applied to tensor powers. We include the proof for completeness in Appendix D.

The *asymptotic rank* of T is

$$\widetilde{R}(T) = \lim_{N\to\infty} \operatorname{rank}(T^{\otimes N})^{1/N} = \inf_{N\geq 1} \operatorname{rank}(T^{\otimes N})^{1/N}.$$

The limit exists because tensor rank is submultiplicative; apply the subadditive-sequence form of Fekete's lemma to its logarithm.

Theorem 3.2 (Strassen's spectral theorem)**.** *For every nonzero tensor T over $\mathbb{C}$,*

$$\widetilde{R}(T) = \max_{\lambda} \lambda(T), \tag{9}$$

where the maximum ranges over all tensor characters.

We use [10, Theorem 3.8]; see also [12, Corollary 3.30 of the author version]. The easy direction follows from $\lambda(T)^N = \lambda(T^{\otimes N}) \leq \operatorname{rank}(T^{\otimes N})$. The theorem supplies the opposite direction. Thus, to prove an upper bound on asymptotic rank, it suffices to bound every character value.

Fix an arbitrary character λ. Its *dot-product parameters* are $p_i = \log_2 \lambda(B_i(2)) \in [0,1]$, for $i = X, Y, Z$. They determine the values at all integer dimensions:

$$\lambda(B_i(q)) = q^{p_i} \qquad (q \geq 1). \tag{10}$$

The interval $[0,1]$ follows from $1 \leq \lambda(B_i(2)) \leq 2$. Tensor powers turn multiplication of dimensions into multiplication of character values; comparison with powers of two then gives (10). Appendix F gives the details. In particular, (3) gives

$$\lambda(\langle A, B, C\rangle) = A^{p_Y} B^{p_Z} C^{p_X}. \tag{11}$$

Our aim is to constrain these three parameters for every character. Define its *convolution profiles* by

$$f_{ij}(r,s) = \lambda(C_{ij}(r,s)). \tag{12}$$

The next counting argument turns the polynomial splittings into inequalities for these profiles.

Suppose T splits into s matched sectors $T_1, \ldots, T_s$ sharing leg i. In $T^{\otimes N}$ a sector is specified by a word of N sector labels. Retain on the two nonshared legs the words with n_h occurrences of label h, where $\sum_h n_h = N$. There are

$$M = \binom{N}{n_1, \ldots, n_s}$$

matched sectors. For example, multiplicities $(2,1)$ give the words $112, 121, 211$. Each sector is a product of the T_h with the prescribed multiplicities. Apply (5); now all legs distinguish the words, so their factors can be reordered independently. This gives

$$(2M)T^{\otimes N} \trianglerighteq M \left(\bigotimes_h T_h^{\otimes n_h} \right) \otimes B_i(M). \tag{13}$$

Lemma 3.3 (Multinomial power-sum inequality)**.** *If $T \trianglerighteq \sum_{h=1}^{s} T_h$ is a splitting into matched sectors sharing leg i and $p_i > 0$, then*

$$\lambda(T)^{1/p_i} \geq \sum_h \lambda(T_h)^{1/p_i}. \tag{14}$$

For fully independent summands, $T \trianglerighteq \bigoplus_h T_h$ gives $\lambda(T) \geq \sum_h \lambda(T_h)$ directly by additivity.

The counting idea is to sum over all multiplicities. With $a_h = \lambda(T_h)^{1/p_i}$, applying the character to (13) and using the multinomial theorem gives

$$\left(\sum_h a_h\right)^N \leq (N+1)^s 2^{1/p_i} \lambda(T)^{N/p_i}.$$

The number of multiplicity tuples is polynomial in N, so it disappears when we take N-th roots and let $N \to \infty$. The full calculation is in Appendix F.

Rank bounds and restrictions give

$$1 \leq f_{ij}(r,s) \leq r + s - 1, \qquad f_{ij}(r,s) = f_{ji}(s,r), \tag{15}$$

and coordinatewise monotonicity. The equality merely interchanges the two polynomial inputs of the same convolution.

Lemma 3.4 (Powered concavity and tripling)**.** *For $p_i > 0$, the sequence $s \mapsto f_{ij}(r,s)^{1/p_i}$ is concave, and for all integers $r, s \geq 1$,*

$$f_{ij}(r, 3s + r - 1)^{1/p_i} \geq 2 f_{ij}(r,s)^{1/p_i} + f_{ik}(r,s)^{1/p_i}, \tag{16}$$

where i, j, k are distinct. For $p_j > 0$, the sequence $r \mapsto f_{ij}(r,s)^{1/p_j}$ is also concave.

For example, the two-sector splitting gives

$$2 f_{ij}(r,s)^{1/p_i} \geq f_{ij}(r, s+1)^{1/p_i} + f_{ij}(r, s-1)^{1/p_i},$$

which is discrete concavity. The three-sector splitting gives (16); input interchange gives concavity in the other coordinate. Appendix F supplies the proof.

4. Rescaling the convolution profiles

We next pass from integer input lengths to real ones, so we can compare convolution shapes at any scale. Replication from Section 2 lets us enlarge both inputs together; the bounds, monotonicity, concavity, and tripling inequalities from the preceding section control the resulting limit. We obtain continuous profiles that retain concavity and tripling and scale linearly when both lengths grow by the same factor.

Applying a character to Lemma 2.2 gives

$$f_{ij}(mr, ms) \geq m f_{ij}(r,s). \tag{17}$$

This makes the rescaled values stable under further replication. For real $r, s > 0$, define

$$g_{ij}(r,s) = \lim_{m \to \infty} \frac{f_{ij}(\lceil mr \rceil, \lceil ms \rceil)}{m}, \qquad m \in \mathbb{Z}_{>0}. \tag{18}$$

The following lemma justifies this limit and lets us work with real lengths throughout the remaining analysis.

Lemma 4.1 (Rescaled profiles)**.** *The limit in* (18) *exists. The profiles* g_{ij} *are continuous, coordinatewise nondecreasing, and satisfy*

$$\min\{r,s\} \le g_{ij}(r,s) \le r+s, \qquad g_{ij}(r,s) = g_{ji}(s,r), \qquad g_{ij}(tr,ts) = t\, g_{ij}(r,s). \tag{19}$$

For $p_i > 0$, g_{ij}^{1/p_i} *is concave in* s *and satisfies*

$$g_{ij}(r,3s+r)^{1/p_i} \ge 2g_{ij}(r,s)^{1/p_i} + g_{ik}(r,s)^{1/p_i}. \tag{20}$$

For $p_j > 0$, g_{ij}^{1/p_j} *is concave in* r*. All lengths and scale factors here may be arbitrary positive reals.*

The discrete inequalities thus become inequalities about homogeneous curves. The additive -1 in (16) disappears at this scale. The proof first compares rational lengths at a common integer scale; homogeneity and monotonicity then give continuity and the real extension. Appendix F includes that argument and verifies that rounding the lengths in (18) does not change the limit.

5. Common slopes and the angular constraint

With the profile properties from Lemma 4.1, we can now relate the three dot-product parameters $p_i \in [0,1]$. Comparing the growth of differently oriented convolutions leads to the angular constraint in Proposition 5.2. This is the only result from this section needed for the running-time bounds in Section 6.

Index the three tensor legs by $1,2,3$. Full proofs of this section's slope and growth claims are in Appendix G.

Lemma 5.1 (Common slopes and affine bounds)**.** *Fix* i *with* $p_i > 0$*, and let* j,k *be the other legs. The functions* $P_{ij}(s) = g_{ij}(1,s)^{1/p_i}$ *and* $P_{ik}(s) = g_{ik}(1,s)^{1/p_i}$ *have a common asymptotic slope* $L_i > 0$*. There are offsets satisfying*

$$c_{ij}, c_{ik} \ge 0, \qquad c_{ij} + c_{ik} \le 1, \tag{21}$$

such that, for all real $s,t > 0$*,*

$$P_{ij}(s+t) \ge P_{ij}(s) + L_i t, \qquad L_i s \le P_{ij}(s) \le L_i(s + c_{ij}),$$

and likewise for P_{ik}*.*

Concavity gives each curve a limiting slope. The coupled tripling inequalities force the two slopes to agree. Writing $R = P_{ij} + P_{ik}$, they also give $R(3s+1) \ge 3R(s)$; iteration yields $R(s) \le L_i(2s+1)$. Consequently the two curves stay above their common limiting line, and their normalized vertical offsets are nondecreasing with sum at most one. Their limits are c_{ij}, c_{ik}. This explains both the affine bounds and the constraint on their sum.

Proposition 5.2 (Angular constraint)**.** *The dot-product parameters of every tensor character satisfy*

$$\sum_{i=1}^{3} \arccos\sqrt{p_i} \ge \frac{\pi}{2}. \tag{22}$$

Proof outline. If some $p_i = 0$, its summand is already $\pi/2$. Assume all three parameters are positive. For this outline, suppose $g = g_{ij}$ is differentiable. The increment and upper bounds in

Lemma 5.1 give $P'_{ij}(t) \geq L_i$ and $P_{ij}(t) \leq L_i(t + c_{ij})$. Homogeneity gives $g(r, s) = rP_{ij}(s/r)^{p_i}$, so, also interchanging the inputs,

$$\frac{\partial_s g(r,s)}{g(r,s)} = \frac{p_i}{r}\frac{P'_{ij}(s/r)}{P_{ij}(s/r)} \geq \frac{p_i}{s + c_{ij}r},$$
$$\frac{\partial_r g(r,s)}{g(r,s)} = \frac{p_j}{s}\frac{P'_{ji}(r/s)}{P_{ji}(r/s)} \geq \frac{p_j}{r + c_{ji}s}.$$

Differentiating $g(tr, ts) = t\, g(r, s)$ at $t = 1$ gives

$$r\frac{\partial_r g(r,s)}{g(r,s)} + s\frac{\partial_s g(r,s)}{g(r,s)} = 1.$$

Substituting the two lower bounds yields

$$1 \geq \frac{p_i s}{s + c_{ij}r} + \frac{p_j r}{r + c_{ji}s} \qquad (r, s > 0). \tag{23}$$

Appendix G obtains the same inequality using finite increments, without assuming differentiability. Clearing denominators gives a quadratic in r/s. Its nonnegativity for every positive ratio implies

$$\sqrt{c_{ij}c_{ji}} \geq \left(\sqrt{p_i p_j} - \sqrt{(1 - p_i)(1 - p_j)}\right)_+, \tag{24}$$

where $(z)_+ = \max\{z, 0\}$.

Put $\theta_i = \arccos\sqrt{p_i} \in [0, \pi/2)$ and $S = \theta_1 + \theta_2 + \theta_3$. Equation (24) and AM–GM give, for each pair,

$$2\cos(\theta_i + \theta_j)\cos\theta_i\cos\theta_j \leq c_{ij}\cos^2\theta_i + c_{ji}\cos^2\theta_j.$$

Summing, using (21), and applying cosine addition gives

$$\begin{aligned} 0 &\geq 2\sum_{i<j}\cos(\theta_i + \theta_j)\cos\theta_i\cos\theta_j - \sum_i \cos^2\theta_i \\ &= \cos S\sum_i \cos\theta_i\cos(\theta_j - \theta_k), \end{aligned} \tag{25}$$

where j, k are the other two indices in the last sum. Every term of that sum is positive, since $\cos\theta_i > 0$ and $|\theta_j - \theta_k| < \pi/2$. Thus $\cos S \leq 0$; as $0 \leq S < 3\pi/2$, we obtain $S \geq \pi/2$. Appendix G supplies the finite-increment argument and the quadratic calculation. □

6. From character bounds to running times

We can now turn the parameter constraint into matrix-multiplication algorithms. The other ingredients are the matrix-tensor formula (11) and Strassen's spectral theorem (Theorem 3.2). For a given matrix shape, we bound a weighted sum of the three parameters for every character; the theorem then bounds asymptotic rank. We first show how this rank bound gives a running-time bound, then apply the angular constraint from Proposition 5.2 to obtain the general-shape bound and the rectangular curve.

Lemma 6.1 (Character bounds give algorithms)**.** *Fix* $\mathbf{h} = (h_X, h_Y, h_Z) \geq 0$*. If the dot-product parameters of every tensor character satisfy* $\sum_i h_i p_i \leq E$*, then* $\omega(h_Y, h_Z, h_X) \leq E$*.*

Proof. The zero target is trivial. For a fixed integer $N \geq 2$, put $D_i = \lceil N^{h_i} \rceil$ and $M_N = \bigotimes_i B_i(D_i)$. Since $D_i \leq 2N^{h_i}$ and $0 \leq p_i \leq 1$, every character satisfies

$$\lambda(M_N) = \prod_i D_i^{p_i} \leq 8N^{\sum_i h_i p_i} \leq 8N^E.$$

Theorem 3.2 therefore gives

$$\widetilde{R}(M_N) \leq 8N^E. \tag{26}$$

Fix any positive exponent slack η. First choose N so large that $8 < N^{\eta/2}$. The infimum defining $\widetilde{R}(M_N)$ then gives a fixed integer K and an exact rank decomposition with

$$R := \operatorname{rank}(M_N^{\otimes K}) \leq N^{K(E+\eta)}.$$

This is a finite bilinear algorithm for matrix dimensions (D_Y^K, D_Z^K, D_X^K). Apply it recursively ℓ times to blocks. By (3), the resulting dimensions are $(D_Y^{K\ell}, D_Z^{K\ell}, D_X^{K\ell})$ and there are R^ℓ scalar multiplications.

Each input and output dimension, viewed as a number of scalar entries, is at most R, by the corresponding matrix-rank lower bound. Thus the linear substitutions and additions over all recursion levels use $O(\ell R^\ell)$ operations. Given input size n, choose $\ell = \lceil \log_N n/K \rceil$ and pad to these dimensions. They cover the target shape because $D_i^{K\ell} \geq n^{h_i}$, while $R^\ell = O(n^{E+\eta})$. This proves the bound for every positive slack, and hence the lemma under our exponent convention. All choices of N, K, and the finite identity precede the limit $n \to \infty$. □

It remains to bound these weighted sums using the angular constraint.

Proof of Theorem 1.1. Reorder the character's dot-product parameters so that the objective is $ap_1 + bp_2 + cp_3$. If $c = 0$, the bound $a + b$ follows from $0 \leq p_i \leq 1$. Assume now $c > 0$ and write $p_i = \cos^2\theta_i$, with $\theta_i \in [0, \pi/2]$. Proposition 5.2 gives $S := \sum_i \theta_i \geq \pi/2$. Scaling every angle by $\pi/(2S)$ only increases the objective. It therefore suffices to bound $W(\theta) = a\cos^2\theta_1 + b\cos^2\theta_2 + c\cos^2\theta_3$ over the relaxed region $\theta_i \geq 0$, $\sum_i \theta_i = \pi/2$.

First take all angles positive. Let $n_i \in \mathbb{R}^2$ be the outward unit normal to the side opposite angle $2\theta_i$ of a triangle with these angles. Then $n_j \cdot n_k = -\cos(2\theta_i)$. Define the coefficients

$$\ell_1 = \sqrt{\frac{bc}{a}}, \qquad \ell_2 = \sqrt{\frac{ac}{b}}, \qquad \ell_3 = \sqrt{\frac{ab}{c}},$$

so $\ell_2\ell_3 = a$, $\ell_3\ell_1 = b$, and $\ell_1\ell_2 = c$. Expanding the squared norm of $v := \sum_i \ell_i n_i$ gives

$$W(\theta) = E(a,b,c) - \frac{1}{4}\|v\|^2, \qquad E(a,b,c) := \frac{a+b+c}{2} + \frac{1}{4}\left(\frac{ab}{c} + \frac{ac}{b} + \frac{bc}{a}\right). \tag{27}$$

In particular, $W(\theta) \leq E(a,b,c)$.

In the thin case $c \leq ab/(a+b)$,

$$\ell_3 - \ell_1 - \ell_2 = \frac{ab - ac - bc}{\sqrt{abc}} \geq 0.$$

The triangle inequality gives $\|v\| \geq \ell_3 - \ell_1 - \ell_2$, so the same identity yields

$$W(\theta) \leq E(a,b,c) - \frac{(ab - ac - bc)^2}{4abc} = a + b.$$

Continuity covers zero angles and degenerate triangles. This proves both branches, including the threshold equality.

We have bounded the objective for every tensor character. Lemma 6.1, with target $\mathbf{h} = (c, a, b)$, gives (1). □

Proof of Corollary 1.2. Apply Theorem 1.1 to the sorted dimensions of $(1, k, 1)$. For $0 \le k \le 1$, the first branch applies exactly when $k \le \frac{1}{2}$. In the remaining range, both possible orderings give

$$\frac{2+k}{2} + \frac{1}{4}\left(2k + \frac{1}{k}\right) = 1 + k + \frac{1}{4k}.$$

The output-size lower bound $\omega(k) \ge 2$ proves equality for $0 \le k \le \frac{1}{2}$ and hence $\alpha \ge \frac{1}{2}$. □

Some values of the bound in Corollary 1.2 are:

k	0.5	0.6	0.8	1	1.5	2
$\omega(k) \le$	2	2.0166…	2.1125	2.25	2.6666…	3.125

The exponent convention gives an $O(n^2)$-time algorithm at $k = \frac{1}{2}$.

7. Applications

Corollary 1.2 and Theorem 1.1 give the following bounds through standard rectangular-multiplication balances. All bounds use arithmetic operations, with graph operations and symbol comparisons also charged.

1. **One lowest common ancestor per pair in a DAG.** Grandoni et al. [4, Theorem 4.2 of the published version] obtain exponent $1 + 2x$ from the balance $\omega(x) = 3x$. Corollary 1.2 permits $x = (1+\sqrt{3})/4$, giving $O(n^{(3+\sqrt{3})/2})$ time for finding one lowest common ancestor for every pair in an n-vertex DAG. The exponent is $(3 + \sqrt{3})/2 \approx 2.36603$.

2. **Exact all-pairs Hamming distance.** For n vectors of length n over an unrestricted alphabet, the reduction to dominance products [7] and Yuster's tradeoff [13, Proposition 2.1] give the balance $\rho = \omega(4 - \rho)$. Corollary 1.2 gives $\rho = (13 - \sqrt{7})/4 \approx 2.58856$ and running time $O(n^\rho)$ for computing every exact Hamming distance.

3. **The standard unweighted APSP balance.** Zwick's deterministic algorithm for directed unweighted APSP [15, Section 6] has exponent $2 + \mu$, where $\omega(\mu) \le 1 + 2\mu$. Here $\omega(\frac{1}{2}) = 2 = 1 + 2(\frac{1}{2})$, so it gives $O(n^{2.5})$ running time. Combining the full curve with the deterministic $(\min, +)$-product algorithm and APSP reduction of Alman and Vassilevska Williams [1, Theorems 22 and 33–34] further gives $O(n^{2.4999})$ running time.

4. **Minimum and maximum witnesses.** For two Boolean $n \times n$ matrices, a minimum (maximum) witness for entry (i, j) is the least (greatest) index k with $A_{ik} = B_{kj} = 1$. The standard blocking algorithm [3, Section 2.1] takes $O(n^{1-r+\omega(r)} + n^{2+r})$ time. Taking $r = \frac{1}{2}$ gives $O(n^{2.5})$ time for all witnesses.

5. **Sparse matrix multiplication.** For two $n \times n$ matrices with at most m nonzero entries each, the sparse–dense decomposition of Yuster and Zwick [14, Section 3] gives $O(m^2/n^r + n^{\omega(r)})$ time for $0 \le r \le 1$. Thus $m \le n^{1.25}$ permits $O(n^2)$ time.

6. **Dynamic matrix inverse and reachability.** Substituting Theorem 1.1 into van den Brand–Nanongkai–Saranurak's tradeoff [11, Theorem 4.2], with $u = 0.50323551$ and $v = 0.80667763$ in $\max\{u + v, \omega(1, u, v) - u, \omega(v) - v\}$, gives $O(n^{1.30992})$ worst-case time per entry update and query for maintaining the inverse of a nonsingular $n \times n$ matrix, after $O(n^{2.25})$ preprocessing. The associated Monte Carlo reduction gives the same bounds for directed reachability under edge updates, with $O(1)$ query time for a fixed pair [11, Corollary C.12 and Theorem C.16].

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Appendix A. Splitting two polynomial products

We recap the two-sector construction used in [9, Lemma 4.1] as an explicit change of coordinates for pairs of polynomials.[1]

Fix $r \geq 1$ and $s \geq 2$. The tensor $C_{XY}(r,s) \otimes B_X(2)$ computes

$$(f,(a,b)) \longmapsto (fa, fb), \qquad \deg f < r, \quad \deg a, \deg b < s.$$

Thus tensoring with $B_X(2)$ means multiplying two polynomials by the same f. We will change coordinates to split the second input into lengths $s+1$ and $s-1$.

For $e \geq 1$ and $\deg a, \deg b \leq e$, use

$$p = a + tb, \qquad h = b - p_{e+1}t^e, \qquad p_d = [t^d]p.$$

Here $[t^d]p$ denotes the coefficient of t^d. We have $\deg p \leq e+1$ and $\deg h \leq e-1$, and these coordinates are invertible: $b = h + p_{e+1}t^e$, $a = p - tb$. Use them on the input pair with $e = s-1$ and on the output pair with $e = r+s-2$. Multiplication by f becomes

$$(p,h) \longmapsto (fp,\ fh + p_s t^{s-1}(f - f_{r-1}t^{r-1})).$$

Scale input h by ε^{-1} and its output coordinate by ε. The map is now

$$(p,h) \longmapsto (fp,\ fh + \varepsilon p_s t^{s-1}(f - f_{r-1}t^{r-1})) \xrightarrow{\varepsilon \to 0} (fp, fh).$$

The independent polynomials p, h have lengths $s+1, s-1$, respectively. This gives $C_{XY}(r, s+1)$ and $C_{XY}(r, s-1)$ sharing the first leg, with disjoint second and third legs. Permuting the legs gives the other orientations.

Appendix B. The three-sector splitting

We recap the construction from [9, Lemma 4.2]. In $C_{XY}(r, 3s+r-1)$, a term has $0 \leq x < r$ and output index $z = x+y$. Partition the second-input and output indices into three intervals:

	left	middle	right
Y	$[0, s-1]$	$[s, 2s+r-2]$	$[2s+r-1, 3s+r-2]$
Z	$[0, s+r-2]$	$[s+r-1, 2s+r-2]$	$[2s+r-1, 3s+2r-3]$.

Scale middle Y variables by ε and middle Z variables by ε^{-1}, leaving the others unchanged. Since $z = y+x$ with $0 \leq x < r$, left and right Y indices stay in the corresponding Z intervals. Cross-sector terms thus connect middle Y to outer Z and acquire a factor ε; matched terms remain unchanged. As $\varepsilon \to 0$, precisely three matched sectors sharing X remain.

Translations identify the outer sectors with $C_{XY}(r,s)$. The middle is

$$\sum_{u=0}^{r-1} \sum_{v=0}^{s-1} x_{r-1-u} y_{s+u+v} z_{s+r-1+v},$$

which is $C_{XZ}(r,s)$ after reversing its first-leg basis. This identifies the individual sector. In the fixed-word construction of (13), the sectors acquire independent first-leg variables before their individual bases are changed.

[1] This is the degree-one case of the classical Clebsch–Gordan formula [6, Theorem 2.6.3, pp. 56–57 of the author version]. The coefficient calculation here does not require that formula.

Appendix C. Finite separation of a shared leg

We adapt the construction of [9, Proposition 3.1] with an equal-sums filter, reducing the copy count from $5M$ to $2M$. It suffices to share leg X. Write

$$T = \sum_{h=1}^{M} \sum_{a,b,c} \mu_{habc} x_a y_{hb} z_{hc}.$$

Take $L = 2M$ copies and a primitive L-th root of unity ζ. In copy $\ell = 0, \dots, L-1$, make the substitutions

$$x_a^{(\ell)} \longmapsto \sum_{g=1}^{M} \zeta^{\ell g} X_{ag},$$

$$y_{hb}^{(\ell)} \longmapsto \sum_{u=1}^{M} \zeta^{\ell(u-h)} Y_{hbu},$$

$$z_{hc}^{(\ell)} \longmapsto \frac{1}{L} \sum_{v=1}^{M} \zeta^{-\ell v} Z_{hcv}.$$

Summing the copies multiplies each coefficient by

$$\frac{1}{L} \sum_{\ell=0}^{L-1} \zeta^{\ell(g+u-h-v)}.$$

The geometric sum is one for exponent divisible by L and zero otherwise. Since $|g+u-h-v| \le 2(M-1) < L$, it retains precisely $g + u = h + v$.

Now give X_{ag}, Y_{hbu}, Z_{hcv} respective weights g^2, $2hu - h^2$, and $-2hv$. On every retained term their total is

$$g^2 + 2h(u - v) - h^2 = (g - h)^2 \ge 0.$$

The weight-zero terms have $g = h$ and $u = v$, so the limit is

$$\sum_{h,a,b,c,u} \mu_{habc} X_{ah} Y_{hbu} Z_{hcu} = \bigoplus_{h} (T_h \otimes B_X(M)).$$

Every leg now distinguishes h. Although individual weights may be negative, the transformed coefficients form a polynomial of degree at most $(M-1)^2$. This proves (5).

Appendix D. Interpolation of a degeneration

For completeness, we give the standard interpolation proof [2, Theorem 8].

Proof of Lemma 3.1. Let $P(\varepsilon)$ be the tensor obtained from T by the degeneration's substitutions. Its coefficients are polynomials in ε of degree at most L, and $P(0) = S$. Thus L describes the chosen degeneration. For every $\varepsilon \neq 0$, $P(\varepsilon)$ is a restriction of T. Put $Q(\varepsilon) = P(\varepsilon)^{\otimes N}$. Each coefficient of Q is a product of N coefficients of P, so its degree is at most NL, and $Q(0) = S^{\otimes N}$.

Choose $NL + 1$ distinct nonzero points $\varepsilon_0, \dots, \varepsilon_{NL}$. Lagrange interpolation, applied to every coefficient, gives

$$S^{\otimes N} = Q(0) = \sum_{j=0}^{NL} c_j Q(\varepsilon_j), \qquad c_j = \prod_{\substack{0 \le h \le NL \\ h \neq j}} \frac{-\varepsilon_h}{\varepsilon_j - \varepsilon_h}.$$

Each $Q(\varepsilon_j)$ is a restriction of $T^{\otimes N}$. To form the displayed weighted sum from their direct sum, identify the corresponding variables on all three legs and scale the variables on one leg of copy j by c_j. These are allowed linear substitutions, proving (8). Applying a character and taking N-th roots gives

$$\lambda(S) \le (NL+1)^{1/N}\lambda(T).$$

Letting $N \to \infty$ proves $\lambda(S) \le \lambda(T)$. No continuity assumption on λ is needed. □

Appendix E. Replicating a convolution

Proof of Lemma 2.2. Put $P(t) = t^m - 1$. Repeated polynomial division writes the inputs uniquely as

$$F = \sum_{a=0}^{r-1} F_a P^a, \qquad G = \sum_{b=0}^{s-1} G_b P^b, \qquad \deg F_a, \deg G_b < m.$$

These are invertible linear coordinates: the basis polynomials $t^d P^a$ have distinct leading degrees $ma+d$. On the output retain the analogous blocks H_c for $0 \le c < r+s-1$. They use $m(r+s-1)$ independent coordinates, within the original output length $m(r+s)-1$. Divide each $F_a G_b = Q_{ab} + PR_{ab}$ with $\deg Q_{ab}, \deg R_{ab} < m$. Scale input blocks a, b by $\varepsilon^{-a}, \varepsilon^{-b}$ and output block c by ε^c. The transformed product has

$$H_c = \sum_{a+b=c} Q_{ab} + \varepsilon \sum_{a+b=c-1} R_{ab}.$$

Let $\alpha_1, \dots, \alpha_m$ be the distinct roots of P. Since $Q_{ab}(\alpha_h) = F_a(\alpha_h)G_b(\alpha_h)$, the limit gives

$$H_c(\alpha_h) = \sum_{a+b=c} F_a(\alpha_h)G_b(\alpha_h) \qquad (1 \le h \le m).$$

Evaluation at m distinct points is invertible on each block of degree less than m. The resulting coordinates therefore give m independent convolutions of lengths r, s. The evaluation step is the usual diagonalization of cyclic convolution [5, Theorem 5 and Section 5.1]; the displayed scalings prove the degeneration, with coefficient degree at most one. □

Appendix F. Counting and rescaling

We give the complete counting argument and justify the real-length profiles used in Sections 3–4.

To prove (10), fix an integer $q \ge 1$ and put $m = \lfloor N \log_2 q \rfloor$. Monotonicity in dimension and $B_i(q)^{\otimes N} \cong B_i(q^N)$ give

$$2^{mp_i} \le \lambda(B_i(q))^N \le 2^{(m+1)p_i}.$$

Take N-th roots and let $N \to \infty$. The interval $[0,1]$ follows from $1 \le \lambda(B_i(2)) \le 2$.

Proof of Lemma 3.3. For every tuple $\sum_h n_h = N$, applying λ to (13) gives

$$2M\lambda(T)^N \ge M^{1+p_i} \prod_h \lambda(T_h)^{n_h}.$$

Put $a_h = \lambda(T_h)^{1/p_i}$. Cancel M and take the power $1/p_i$:

$$M \prod_h a_h^{n_h} \le 2^{1/p_i} \lambda(T)^{N/p_i}.$$

There are at most $(N+1)^s$ multiplicity tuples. Summing and using the multinomial theorem gives

$$\left(\sum_h a_h\right)^N \le (N+1)^s 2^{1/p_i} \lambda(T)^{N/p_i}.$$

For fixed s, this extra factor is polynomial in N. Taking N-th roots and letting $N \to \infty$ proves (14). □

Proof of Lemma 3.4. The two-sector splitting and Lemma 3.3 give, for $s \ge 2$,

$$2 f_{ij}(r,s)^{1/p_i} = \lambda(C_{ij}(r,s) \otimes B_i(2))^{1/p_i} \ge f_{ij}(r,s+1)^{1/p_i} + f_{ij}(r,s-1)^{1/p_i}.$$

This is discrete concavity in s; input interchange gives the other coordinate. The three-sector splitting and the same lemma give (16). □

Proof of Lemma 4.1. Initially let $r, s > 0$ be rational, and define

$$g_{ij}(r,s) = \sup_{\substack{q \in \mathbb{Q}_{>0} \\ qr, qs \in \mathbb{Z}_{>0}}} \frac{f_{ij}(qr, qs)}{q}. \tag{F.1}$$

Replication gives $f_{ij}(m,m) \ge m$, and monotonicity therefore gives $f_{ij}(r,s) \ge \min\{r,s\}$ for integer lengths. The bounds and input interchange follow from (15) and the lower bound above. Replacing q by q/t proves homogeneity.

The crucial point is that finitely many supremum values can be approached at one common scale. Choose scales q_h approximating the desired values, and choose Q so that every Q/q_h is a positive integer. By (17),

$$\frac{f_{ij}(Qr, Qs)}{Q} \ge \frac{f_{ij}(q_h r, q_h s)}{q_h}.$$

Replacing Q by an integer multiple preserves all these approximations and can clear any additional rational denominators. Monotonicity now follows by comparing lengths at a common scale.

For concavity, take $s = \tau s_1 + (1-\tau)s_2$, with rational $0 \le \tau \le 1$. Approach $g_{ij}(r, s_1)$ and $g_{ij}(r, s_2)$ at one scale Q, with Qr, Qs_1, Qs_2, Qs integers. Discrete concavity gives

$$\left(\frac{f_{ij}(Qr, Qs)}{Q}\right)^{1/p_i} \ge \tau \left(\frac{f_{ij}(Qr, Qs_1)}{Q}\right)^{1/p_i} + (1-\tau)\left(\frac{f_{ij}(Qr, Qs_2)}{Q}\right)^{1/p_i}.$$

Bound the left side by $g_{ij}(r,s)^{1/p_i}$ and let the approximation error tend to zero. Input interchange gives the other coordinate.

For tripling, monotonicity first enlarges the discrete source length $3s + r - 1$ to $3s + r$. At a common scale Q,

$$\left(\frac{f_{ij}(Qr, Q(3s+r))}{Q}\right)^{1/p_i} \ge 2\left(\frac{f_{ij}(Qr, Qs)}{Q}\right)^{1/p_i} + \left(\frac{f_{ik}(Qr, Qs)}{Q}\right)^{1/p_i}.$$

Approach the two right-hand suprema simultaneously; the left side is bounded by $g_{ij}(r, 3s + r)^{1/p_i}$. This proves (20).

To extend the profiles to real lengths, we first compare two positive rational pairs (r,s) and (r',s'). Put

$$a = \min\left\{\frac{r'}{r}, \frac{s'}{s}\right\}, \qquad b = \max\left\{\frac{r'}{r}, \frac{s'}{s}\right\}.$$

Homogeneity and monotonicity give

$$a\, g_{ij}(r,s) \le g_{ij}(r',s') \le b\, g_{ij}(r,s).$$

Together with $g_{ij}(r,s) \le r+s$, this implies

$$|g_{ij}(r',s') - g_{ij}(r,s)| \le \frac{r+s}{\min\{r,s\}} \max\{|r'-r|, |s'-s|\}. \tag{F.2}$$

The coefficient is bounded on every closed bounded region that stays away from the coordinate axes. Approximating a positive real pair by rationals therefore gives a unique continuous extension. All the rational bounds, homogeneity, monotonicity, input interchange, and powered concavity extend by continuity. So does (20).

We finally check that this extension agrees with the limit in (18). For positive integers A, B, the scale $q = 1$ in (F.1) gives $f_{ij}(A,B) \le g_{ij}(A,B)$. Thus for arbitrary real $r, s > 0$,

$$\frac{f_{ij}(\lceil mr \rceil, \lceil ms \rceil)}{m} \le g_{ij}\left(\frac{\lceil mr \rceil}{m}, \frac{\lceil ms \rceil}{m}\right) \longrightarrow g_{ij}(r,s),$$

using homogeneity and continuity.

For the reverse bound, fix positive rationals $r_0 < r$, $s_0 < s$ and an admissible scale q with qr_0, qs_0 positive integers. Put $k = \lfloor m/q \rfloor$. For all sufficiently large m, $k \ge 1$, and $kqr_0 \le \lceil mr \rceil$, $kqs_0 \le \lceil ms \rceil$. Monotonicity and replication give

$$\frac{f_{ij}(\lceil mr \rceil, \lceil ms \rceil)}{m} \ge \frac{k}{m} f_{ij}(qr_0, qs_0).$$

Since $k/m \to 1/q$, taking the lower limit and then the supremum over q gives a lower bound $g_{ij}(r_0, s_0)$. Let r_0, s_0 approach r, s from below and use continuity. This proves the limit formula (18), and completes the lemma. □

Appendix G. Common slopes and the angular constraint

Here we justify the common slopes and offsets, then derive the angular constraint using finite increments rather than derivatives.

Proof of Lemma 5.1. Write $P = P_{ij}$ and $Q = P_{ik}$. For $s > 1$, the nonnegative secant slope $(P(s) - P(1))/(s-1)$ decreases as s grows, by concavity. It therefore has a finite limit, which is also $L_P = \lim P(s)/s$. Likewise define $L_Q = \lim Q(s)/s$. Dividing the coupled inequalities

$$P(3s+1) \ge 2P(s) + Q(s), \qquad Q(3s+1) \ge 2Q(s) + P(s)$$

by s and passing to infinity gives $L_P \ge L_Q$ and $L_Q \ge L_P$. Call their common value L_i.

Their sum $R = P + Q$ satisfies $R(3s+1) \ge 3R(s)$. Iterating at $s_m = 3^m(s + \frac{1}{2}) - \frac{1}{2}$ gives

$$R(s) \le \lim_{m\to\infty} 3^{-m} R(s_m) = L_i(2s+1). \tag{G.1}$$

Since $R(s) > 0$, we have $L_i > 0$. For $v > u > 0$, concavity bounds the secant slope $(P(v) - P(u))/(v-u)$ below by every later secant slope; sending its far endpoint to infinity gives

$$P(v) - P(u) \ge L_i(v-u).$$

Letting $u \downarrow 0$ and using $P(u) \ge 0$ gives $P(v) \ge L_i v$. Thus the two residuals $P(s)/L_i - s$ and $Q(s)/L_i - s$ are nonnegative and nondecreasing, with sum at most one by (G.1). Both therefore have finite limits; define

$$c_{ij} := \lim_{s\to\infty} \left(\frac{P(s)}{L_i} - s\right), \qquad c_{ik} := \lim_{s\to\infty} \left(\frac{Q(s)}{L_i} - s\right).$$

Taking limits gives $c_{ij} + c_{ik} \le 1$, and bounding each residual by its limit gives the affine upper bounds. □

Proof of Proposition 5.2. If some $p_i = 0$, its summand is already $\pi/2$. Assume all three parameters are positive. Lemma 5.1 gives six offsets and their three bounds

$$c_{ij} \ge 0, \qquad c_{ij} + c_{ik} \le 1.$$

The increment and affine bounds give

$$\frac{P_{ij}(s+t)}{P_{ij}(s)} \ge 1 + \frac{L_i t}{P_{ij}(s)} \ge 1 + \frac{t}{s + c_{ij}}.$$

Using homogeneity, we obtain

$$\frac{g_{ij}(r, s+t)}{g_{ij}(r,s)} \ge \left(1 + \frac{t}{s + c_{ij} r}\right)^{p_i}. \tag{G.2}$$

Increase r by εr, then s by εs. Input interchange supplies the first bound and (G.2) supplies the second:

$$\begin{aligned} 1 + \varepsilon &= \frac{g_{ij}((1+\varepsilon)r, (1+\varepsilon)s)}{g_{ij}(r,s)} \\ &\ge \left(1 + \frac{\varepsilon r}{r + c_{ji} s}\right)^{p_j} \left(1 + \frac{\varepsilon s}{s + c_{ij}(1+\varepsilon) r}\right)^{p_i}. \end{aligned}$$

Take logarithms, divide by ε, and let $\varepsilon \downarrow 0$. The elementary limit $\log(1 + \varepsilon x)/\varepsilon \to x$ gives

$$1 \ge \frac{p_i s}{s + c_{ij} r} + \frac{p_j r}{r + c_{ji} s} \qquad (r, s > 0).$$

With $t = r/s$, this is equivalent to

$$(1 - p_j) c_{ij} t^2 + (1 - p_i - p_j + c_{ij} c_{ji}) t + (1 - p_i) c_{ji} \ge 0 \qquad (t > 0).$$

A quadratic $At^2 + Bt + C$ with $A, C \ge 0$ is nonnegative on $t > 0$ exactly when $B \ge -2\sqrt{AC}$. Indeed, divide by t and use $\inf_{t>0}(At + C/t) = 2\sqrt{AC}$; if A or C is zero, the infimum is zero, approached at an endpoint. Thus writing $x = \sqrt{c_{ij} c_{ji}}$ gives $\left(x + \sqrt{(1-p_i)(1-p_j)}\right)^2 \ge p_i p_j$, or

$$\sqrt{c_{ij} c_{ji}} \ge \left(\sqrt{p_i p_j} - \sqrt{(1-p_i)(1-p_j)}\right)_+,$$

where $(z)_+ = \max\{z, 0\}$.

Put $\theta_i = \arccos \sqrt{p_i} \in [0, \pi/2)$ and $S = \theta_1 + \theta_2 + \theta_3$. The pairwise bounds and AM–GM give

$$2 \cos(\theta_i + \theta_j) \cos\theta_i \cos\theta_j \le c_{ij} \cos^2\theta_i + c_{ji} \cos^2\theta_j.$$

Summing over pairs and using the three row budgets gives

$$2 \sum_{i<j} \cos(\theta_i + \theta_j) \cos\theta_i \cos\theta_j \le \sum_i \cos^2\theta_i.$$

For the other two indices j, k, cosine addition gives the identity

$$\cos(\theta_i + \theta_j) \cos\theta_j + \cos(\theta_i + \theta_k) \cos\theta_k - \cos\theta_i = \cos S \cos(\theta_j - \theta_k).$$

Multiplying by $\cos\theta_i$ and summing factors the preceding nonpositive difference as

$$0 \ge \cos S \sum_i \cos\theta_i \cos(\theta_j - \theta_k).$$

Every term in the sum is positive because $\theta_i < \pi/2$ and $|\theta_j - \theta_k| < \pi/2$. Hence $\cos S \le 0$. Since $0 \le S < 3\pi/2$, this gives $S \ge \pi/2$ as required. □